Ensemble Learning for Face Identification on the Olivetti Faces Dataset: Decision Trees, Bootstrap Aggregation, and Multiclass SAMME Boosting

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GitHub Repository: github.com/Naina220303/ML-Assignment-2 | **Final Commit:** `404fa2f`

Abstract

We compare a tuned decision tree, bootstrap aggregation (bagging), and multiclass SAMME boosting for face identification on the Olivetti Faces dataset, using raw pixels and a fixed identity-stratified split. A 200-tree bagging ensemble reaches 82.5% test accuracy versus 52.5% for the selected tree, and keeps 68.75% accuracy when a central 16×16 region is zeroed. SAMME is less consistent. Metrics are transcribed from the packaged notebook outputs and have not yet been verified by an independent clean rerun.

1 Introduction

Face identification assigns an input face image to one of a known set of identities. Modern systems rely on learned representations and deep networks, but classical classifiers remain useful for understanding model capacity, data variability, and ensemble effects. This assignment compares a decision tree, bootstrap aggregation (bagging), and multiclass SAMME boosting.

A decision tree recursively partitions feature space and provides an interpretable baseline. Bagging trains trees on bootstrap-resampled datasets and aggregates their predictions, primarily reducing variance. SAMME extends AdaBoost to multiclass classification by reweighting difficult training examples and combining weak learners through weighted voting.

Contributions. This report makes the following contributions:

- a validation-tuned single-tree baseline on a fixed identity-stratified split;
- a bagging study over ensemble sizes $B \in \{1, \ldots, 200\}$, including out-of-bag, disagreement, and vote-margin diagnostics;
- a multiclass SAMME boosting comparison with a worked toy example;
- a central-occlusion stress test and a repeated-resampling bias–variance and depth analysis.

Section 2 describes the data, Sections 3–5 the preprocessing and methods, Section 6 the protocol, Section 7 the results, and Sections 8–9 the limitations and conclusions. Section 10 collects the detailed logs, and Section 11 the reproducibility checklist.

2 Dataset and Experimental Split

The experiments use the Olivetti Faces collection from scikit-learn: 400 grayscale images of 40 identities, 10 per identity. Each 64×64 image is flattened

into a 4,096-dimensional vector. The task is closed-set identification over 40 classes.

A deterministic per-identity split uses NumPy's default random generator with seed 26011. For each identity, six images go to training, two to validation, and two to testing, giving 240, 80, and 80 images. The split indices are stored in `olivetti_split.npz`.

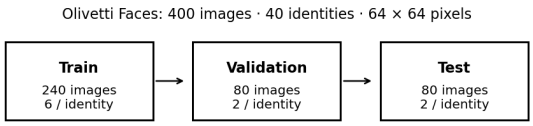


Figure 1: Identity-stratified data partition.

3 Preprocessing and Controlled Occlusion

No learned feature extractor is used. Each image is flattened directly into $x \in \mathbb{R}^{4096}$. This raw-pixel representation is simple to implement but sensitive to pose, expression, illumination, and alignment.

To assess sensitivity to missing local information, rows 24–39 and columns 24–39 of each test image are set to zero. This central 16×16 mask is applied to test images only, in the clean-versus-masked comparisons. It is a controlled stress test, not a complete model of real-world occlusion.



Figure 2: Example of central image masking.

4 Learning Methods

4.1 Decision Tree

The notebook searches maximum depths $\{2,4,8,16,\texttt{None}\}$ and minimum leaf sizes $\{1,2,4\}$, and selects the candidate with the highest validation accuracy. The recorded search selects an unrestricted tree with minimum leaf size one, which obtains 1.00 training, 0.60 validation, and 0.525 test accuracy, a substantial generalization gap.

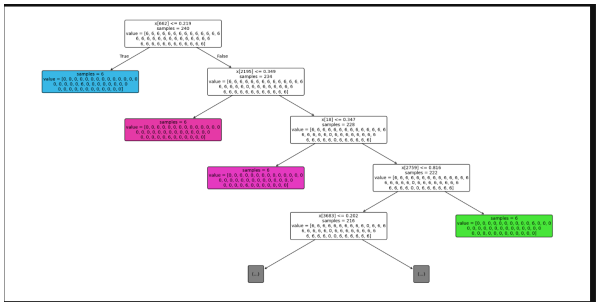


Figure 3: Selected decision tree, truncated to four levels for readability.

4.2 Bagging

For each of B estimators, a bootstrap dataset is formed by sampling N training observations with replacement. A tree is fitted to each sample and predictions are combined by majority vote:

$$h_{\text{bag}}(x) = \text{mode}\{h_1(x), \ldots, h_B(x)\}. \qquad (1)$$

The notebook evaluates $B \in \{1,5,10,25,50,100,200\}$ using the selected tree parameters and seed 26011.

4.3 Out-of-Bag and Diversity Diagnostics

Observations omitted from a bootstrap sample form that estimator's out-of-bag (OOB) set. The notebook also computes pairwise disagreement, i.e. how

often two classifiers predict different labels, and mean vote margin, i.e. the normalized difference between the most- and second-most-supported classes.

5 Multiclass SAMME Boosting

SAMME maintains a weight distribution over training observations. At each round a weak classifier is fitted using the current weights, and misclassified observations receive increased weight. For weighted error ε_t and K classes, the estimator coefficient is

$$\alpha_t = \log \frac{1-\varepsilon_t}{\varepsilon_t} + \log(K-1). \tag{2}$$

The final prediction is a weighted vote. The notebook searches weak-learner depths 1–4 and requested rounds 10, 25, 50, and 100. The best recorded setting is depth four with 100 rounds: training accuracy 0.9708, validation accuracy 0.475, and test accuracy 0.425.

A nine-sample, three-class toy example illustrates the update: the weighted error is $1/3$, so $\alpha = \log 2 + \log 2 = \log 4 \approx 1.3863$, and incorrect examples receive increased weight. The weaker generalization relative to bagging may reflect sensitivity to difficult or ambiguous examples; this is a possible interpretation, not a causal result.

6 Experimental Protocol

Hyperparameters are selected on validation data. Clean and masked comparisons use the held-out test partition. Q3(a) uses five stratified resamples with seed 26011. Q3(b) uses three resamples with seed 26012; Q3(c) and Q3(d) reuse the Q3(b) resamples.

Bias and variance are estimated from predictions across resampled training sets using the notebook's implementation. Because classifier decompositions depend on the chosen loss and estimator, these quantities are treated as the notebook's reported metrics rather than a universal zero-one-loss decomposition.

7 Results

Table 1 summarizes the main outcomes; the subsections below give the detail.

Model (Q1)	Val.	Test	Masked
Decision tree (unrestricted)	0.600	0.525	0.275
Bagging, B=200	0.900	0.825	0.688
SAMME (depth 4, 100 rounds)	0.475	0.425	–

Table 1: Headline accuracy of the three methods. SAMME's masked score comes from a separate Q2(d) experiment (0.400) and is not directly comparable.

7.1 Decision Tree and Bagging

Validation accuracy of the single tree increases from 0.075 at depth two to 0.60 for an unrestricted tree; the latter reaches perfect training accuracy, suggesting overfitting.

Bagging test accuracy increases from 0.35 with one tree to 0.825 with 200 trees (Fig. 4, Table 3). Validation accuracy peaks at 0.9125 for $B = 100$ and is 0.90 at $B = 200$, so it is not strictly monotonic.

From $B = 5$ to $B = 200$, pairwise disagreement decreases from 0.8225 to 0.7905 while mean vote margin increases from 0.2425 to 0.3020 (Fig. 5). The ensemble stays diverse while producing somewhat more decisive aggregate votes.

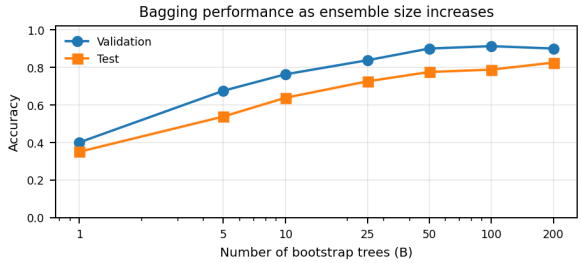


Figure 4: Validation and test accuracy as ensemble size B increases.

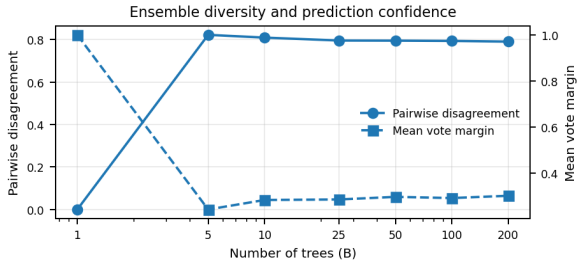


Figure 5: Pairwise disagreement and mean vote margin versus B.

7.2 Central Masking

The selected tree falls from 0.525 clean accuracy to 0.275 masked; bagging falls from 0.825 to 0.6875. The drop is 25 percentage points for the tree and 13.75 for bagging (Fig. 6).

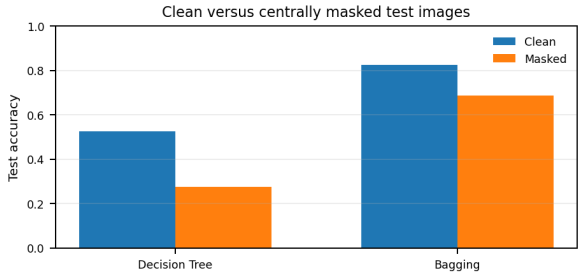


Figure 6: Clean and masked test accuracy for the selected tree and bagging.

In a separate Q2(d) setup, bagging obtains 0.55 clean and 0.3875 masked accuracy; SAMME obtains 0.425 and 0.40; and a depth-four tree obtains 0.1125 in both conditions. These are distinct experiments and should not be merged with Q1(d).

7.3 Bias–Variance and Depth

For Q3(a), bagging reduces reported variance from 0.5570 for a single tree to 0.0458, while mean test accuracy rises from 0.455 to 0.67. SAMME reports low variance (0.0253), high bias squared (0.9082), and mean accuracy 0.16 (Fig. 7).

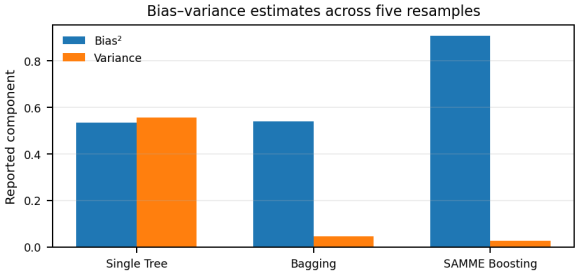


Figure 7: Reported bias-squared and variance estimates over five resamples.



Figure 8: Single-tree and bagging accuracy across depth settings.

In Q3(b), bagging accuracy is 0.2917 at depth two, 0.6250 for unrestricted depth, and 0.6417 at depth 32; the largest variance reduction is 0.4207 for unrestricted depth. In Q3(c), the accuracy improvement peaks at 0.2083 for depth 16, and mean leaf occupancy decreases from 80 at depth two to 6 for an unrestricted tree (Fig. 8).

8 Limitations

The dataset is small and controlled; raw pixels are sensitive to alignment and illumination; only one central mask is evaluated; and resampling counts are limited. Q2 and Q3 experiments use different seeds and protocols. The Q1(b) log's "average distinct samples reported" column is constant at 10 and is not a reliable bootstrap-uniqueness statistic. The CSV artifacts were assembled from recorded notebook outputs rather than regenerated by a clean execution; a fresh rerun and exact dependency lock are required before claiming full reproducibility.

9 Conclusion

Among the evaluated methods, bagging performs best overall: it improves test accuracy over the selected tree, reduces reported variance, and retains more accuracy under central occlusion. SAMME demonstrates sequential reweighting but generalizes less effectively in the recorded configuration. The findings highlight the interaction between learner capacity, sample size, diversity, and training protocol. Future work should rerun from a clean environment, increase the number of resamples, evaluate multiple occlusion patterns, and keep the original notebook intact, fixing execution issues in a separate reproduction copy.

10 Detailed Experimental Logs

The tables below preserve the principal numerical results recorded in the notebook artifacts so comparisons can be audited without relying only on plots. Values are rounded to four decimals where appropriate.

10.1 Decision-Tree Search

Max depth	Min leaf	Train acc.	Val. acc.
2	1	0.0750	0.0750
4	1	0.1250	0.1000
8	1	0.2250	0.1625
16	1	0.4250	0.3125
None	1	1.0000	0.6000

Table 2: Decision-tree hyperparameter search (validation-selected row: unrestricted).

10.2 Bagging Ensemble-Size Sweep

Trees	Train acc.	Val. acc.	Test acc.
1	0.7583	0.4000	0.3500
5	0.9750	0.6750	0.5375
10	1.0000	0.7625	0.6375
25	1.0000	0.8375	0.7250
50	1.0000	0.9000	0.7750
100	1.0000	0.9125	0.7875
200	1.0000	0.9000	0.8250

Table 3: Bagging accuracy versus number of trees.

The Q1(b) artifact also has a distinct-sample column that is constant at 10. Because the notebook's bookkeeping variable is not the bootstrap index array, this column is not treated as a valid estimate of bootstrap uniqueness.

10.3 Ensemble Diagnostics

The recorded OOB accuracy is 0.3266. It is an internal diagnostic and is not interchangeable with held-out test accuracy.

Trees	Pairwise disagreement	Mean vote margin
1	0.0000	1.0000
5	0.8225	0.2425
10	0.8094	0.2838
25	0.7958	0.2860
50	0.7953	0.2973
100	0.7939	0.2921
200	0.7905	0.3020

Table 4: Diversity and confidence diagnostics.

10.4 Clean and Masked Evaluation

Q1(d) model	Clean test	Masked test
Selected decision tree	0.5250	0.2750
200-tree bagging	0.8250	0.6875
Q2(d) model	Clean test	Masked test
SAMME	0.4250	0.4000
Depth-4 tree	0.1125	0.1125
100-tree bagging	0.5500	0.3875

Table 5: Clean vs. masked test accuracy. Q1(d) and Q2(d) are separate experiment blocks and should not be pooled into one ranking.

10.5 Bias–Variance and Depth Results

Q3(a) model	Bias^2	Var.	Mean acc.	Std. acc.
Single tree	0.5330	0.5570	0.4550	0.0232
Bagging	0.5407	0.0458	0.6700	0.0257
SAMME	0.9082	0.0253	0.1600	0.0682

Table 6: Bias–variance estimates over five resamples.

	Single tree			Bagging			
Depth	Bias^2	Var.	Acc.	Bias^2	Var.	Acc.	Var. removed
2	0.9233	0.0359	0.0667	0.9301	0.0018	0.2917	0.0341
None	0.6500	0.4583	0.4458	0.5550	0.0376	0.6250	0.4207
32	0.6564	0.4162	0.4167	0.5625	0.0344	0.6417	0.3818

Table 7: Q3(b) bias–variance and accuracy by tree depth.

Depth	2	4	8	16	None
Accuracy gain	0.0000	0.0333	0.1542	0.2083	0.1958
Leaf occupancy	80.0	48.0	25.8	13.6	6.0
Disagreement	0.9513	0.9434	0.9324	0.8924	0.8163

Table 8: Q3(c) results by tree depth.

10.6 Interpretation of the Complete Logs

Bagging shows the clearest benefit when the base trees are expressive enough to fit different bootstrap samples. In Q3(b), the variance reduction is small at depth two because the shallow learner already has low variance, whereas deeper trees show a much larger reduction. The Q3(c) gain peaks at depth 16 rather than at unrestricted depth, so maximizing individual tree complexity does not necessarily maximize ensemble benefit.

SAMME should be interpreted separately. Its low reported variance, high bias-squared, and low mean accuracy in Q3(a) suggest stable but systematically inaccurate predictions under the notebook's metric and configuration. A stronger follow-up would vary the number of repetitions, examine class-wise confusion, and check whether the boosting implementation and stopping conditions suit this small, high-dimensional training set.

10.7 Further Methodological Interpretation

A bootstrap sample has the same nominal size as the training set, but repeated draws mean some observations occur more than once and others are omitted. The omitted observations form the OOB subset, and aggregating OOB predictions gives an internal estimate based only on training data. The recorded OOB accuracy of 0.3266 is appreciably lower than the test accuracy of the largest ensemble. This can occur because OOB predictions come from varying subsets of the ensemble with fewer votes per observation; it should prompt inspection of OOB coverage and implementation details rather than be treated as a contradiction.

The vote margin is a complementary diagnostic. If the most frequent class receives vote fraction $p_{(1)}$ and the runner-up $p_{(2)}$, the margin is $p_{(1)} - p_{(2)}$. A larger margin means the leading class has more support relative to its nearest competitor, but it is not a calibrated probability and does not guarantee correctness. The rise from 0.2425 at five trees to about 0.3020 at 200 trees indicates somewhat more decisive aggregate decisions.

The toy SAMME example clarifies why boosting behaves differently from bagging. For a weak learner with weighted error $1/3$ in a three-class problem,

$$\alpha = \log \frac{1-1/3}{1/3} + \log(3-1) = \log 4. \qquad (3)$$

The update increases the weights of misclassified examples and decreases the relative weights of correct ones, so later learners see a changed training distribution. Bagging does not adapt its sampling to earlier mistakes; its trees are fitted independently and aggregated afterward. This is why bagging often stabilizes high-variance trees, while boosting may be more sensitive to hard examples and label noise.

The masking protocol has an interpretation boundary: the same central square is applied deterministically to every test image, isolating one form of spatial information loss. It does not test robustness to masks at arbitrary locations, scales, or shapes. A follow-up should sample masks at multiple positions and sizes and report performance by identity and by occlusion severity, to reveal whether the ensemble's gain is broadly distributed or driven by a subset of identities.

Finally, comparisons should respect their experimental blocks. Q1(d), Q2(d), Q3(a), and Q3(d) use different model choices, seeds, and resampling protocols; ranking absolute values across blocks would conflate changes in configuration with changes in evaluation design. The tables therefore retain the original question labels.

10.8 Threats to Validity and Recommended Extensions

Sample size. The training set has 240 images but 4,096 pixel coordinates. A flexible tree can exploit accidental correlations, and a bootstrap ensemble cannot create new independent information. The bagging gain should be read as reduced estimator instability on this fixed benchmark, not as evidence that the method solves unconstrained face recognition.

Split granularity. With two validation and two test images per identity, one image changes per-class accuracy by 50 percentage points. Overall accuracy hides identity-level variation; a stronger

evaluation would report a confusion matrix, macro-averaged precision and recall, per-identity accuracy, and repeated stratified cross-validation or multiple independent partitions.

Decomposition definition. Bias-squared and variance depend on the prediction representation and the decomposition implemented in the notebook. For multiclass predictions, probability-vector variance, class-wise variance, and zero-one-loss variance are not identical, so values should not be compared numerically with studies that use a different loss or estimator.

Software dependence. Scikit-learn defaults and API behavior can change across releases, and tie-breaking can affect trees and votes. Reproducibility requires recording Python, NumPy, and scikit-learn versions, preserving the split indices, and rerunning from a clean kernel. The configuration JSON is a human-readable record and is not loaded automatically by the notebook.

Extensions. Useful next steps include comparing raw pixels against PCA and a learned face representation on the same partitions to separate the effect of aggregation from the limits of raw pixels, plus random-forest feature subsampling, calibrated confidence estimates, multiple occlusion patterns, and confidence intervals over more resamples. These are proposed future work, not reported results.

11 Reproducibility and Artifact Checklist

The repository contains the original notebook, extracted experiment configuration, requirements file, CSV logs, and figure artifacts. The report bundle includes the PDF, editable LaTeX source, figures, and captured result logs. A clean reproduction should: (i) create the documented Python environment; (ii) install the listed dependencies; (iii) execute the notebook from a fresh kernel; (iv) preserve the generated split indices; (v) export the executed notebook separately; and (vi) compare the resulting metrics with the tables above. Exact package versions should be captured with `pip freeze`. The original notebook code remains the source of truth and has not been edited for this report.

References

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